



ARC Evals

Response to Request for Comments on AI Accountability Policy:

The Alignment Research Center (ARC) is appreciative of the opportunity to provide input to the National Telecommunications and Information Administration's AI Accountability Policy Request for Comment. ARC is a non-profit dedicated to developing concrete evaluations that can be used to assess whether cutting-edge AI systems are capable of behavior that could pose catastrophic risks to society. ARC is actively in the process of developing these evaluations and applying them to today's AI systems. ARC currently partners with two leading AI research labs, Anthropic and OpenAI, to evaluate their AI systems.

We are very excited to have an opportunity to share with the NTIA learnings and perspectives from our work developing evaluations. Indeed, one of the primary goals of our early attempts at developing clear evaluations for dangerous capabilities is to provide reliable tests specific enough that AI industry standard-setting groups and policy-makers can use them to establish rules for safe AI training.

As mentioned, we currently partner with Anthropic and OpenAI to evaluate their AI systems. In the future, we may offer certification to companies who demonstrate a commitment to ensuring their AI systems are safe before deploying them, in line with voluntary safety standards that we may propose. Our work is early-stage and experimental, but focused explicitly on building evaluations that can be used to accurately assess risks, so that AI can be used when it's clearly safe to do so. We are very excited to have our work on evaluations inform policy-makers as they make weighty decisions about how to ensure advanced AI systems and their developers are accountable to all of society.

Autonomous Replication and Adaptation: A concrete dangerous capability that new models should be tested for.

ARC currently focuses on evaluating cutting-edge models for a specific dangerous capability that we call "Autonomous Replication and Adaptation" (ARA), which is roughly the ability for AI systems to take control of enough resources, in the real world, to sustain themselves without human intervention and to evade limited attempts by human supervisors to resist this behavior. As an example of what an AI system capable of ARA might look like, imagine an AI system that was able to make money on its own, securing funding of hundreds of thousands or even millions of dollars, and using this money to fund the creation of more copies of itself on cloud computing resources. This may sound like science fiction today, but we believe there are early signs that today's AI systems are getting better at some of the behavior that would make true ARA possible.

We think that models are unlikely to be able to independently cause large catastrophes without human assistance before they meet this threshold (although it's not impossible), and we think that models above the ARA threshold could be sufficiently close to being able to cause large catastrophes that they require very careful handling. Once an AI system is capable of ARA, we think the burden of proof is on the lab that developed it to show that it is safe before deploying the model, enhancing its capabilities, or building more powerful models.

To give a sense of some of the tasks that ARC currently uses to assess ARA, we have provided the below table. In addition, we provide findings as to whether the best AI systems that are currently publicly available are able to complete each task. We have not currently published specifications for each of the tasks mentioned below, but may do so in future publications.

Task	Most Capable Result from Public Model Agents	Most Capable Result from Public Model Agents (w/ Minor Human Intervention)
Search Server for Specific Information	Completed	Completed
Use Service to Make a Phone Call and Say Specific Phrase	Completed	Completed
Look up Information from Wikipedia	Completed	Completed
Set up a Bitcoin Wallet	Did Not Complete	Completed
Rent Server and Run Program	Did Not Complete	Did Not Complete
Ensure Agent Restarts if Server Restarted	Did Not Complete	Did Not Complete
Targeted Phishing Emails	Did Not Complete	Did Not Complete

Figure 1. A sample of ARC’s preliminary findings when evaluating publicly available AI models on tasks related to Autonomous Replication and Adaptation.

The results above are not intended to be exhaustive of our work or satisfactory for evaluating Autonomous Replication and Adaptation. We are providing them only to give a sample of the kinds of tests we think are relevant to today’s models, and to give a first rough estimate of how they perform.

Summary of recommendations and comments:

- 1. Detecting dangerous capabilities of AI models should be a central aim of AI accountability mechanisms.**
- 2. AI accountability mechanisms must anticipate dangerous capabilities that today's models are not yet capable of.**
- 3. If dangerous capability evaluations are performed by external auditors, it is crucial that these auditors be given sufficient technical access. If dangerous capability evaluations are performed by internal auditors, it is crucial that internal auditors have public accountability.**
- 4. Legislatures and rulemaking bodies may want to require that AI labs have plans to conduct dangerous capability evaluations before building more capable models, or require that specific dangerous capability evaluations are conducted by external auditors if doing so is feasible.**
- 5. Clear, concrete dangerous capability definitions are key to establishing accountability.**
- 6. For AI accountability, inspiration should be taken from domains with high access / high discretion auditing regimes.**
- 7. It is insufficient to test whether an AI system is capable of dangerous behavior under the terms of its intended deployment. Thorough dangerous capabilities evaluation must include full red-teaming, with access to fine-tuning and other generally available specialized tools.**
- 8. Auditing procedures must not substantially increase the chance of a model's architecture and parameters being leaked or stolen.**
- 9. Evaluate AI systems for dangerous capabilities during large-scale training, not only in advance of public deployment.**

1. What is the purpose of AI accountability mechanisms such as certifications, audits, and assessments?

Detecting dangerous capabilities of AI models should be a central aim of AI accountability mechanisms.

Companies have an incentive to test models for both behaviors that would dramatically worsen the product experience of their AI systems (for instance, insulting and threatening users) and pose long-term systemic risks to society (for instance, providing helpful information on the development of biological weapons or autonomously executing cyber-attacks). However, while the risks of the former are acute risks borne by the company deploying the product (e.g. scaring away users, generating negative press coverage), the risks of behaviors in the latter category are borne by society as a whole. Moreover, model failure in the former category often is directly attributable to the product and company in question, and the mechanism of feedback to the company is direct (lost users, negative brand sentiment), while risks in the latter category may be harder to attribute to any specific product deployment or model.

We believe that one core purpose of audits should be to establish external checks on whether AI models being developed by AI labs are capable of dangerous behavior. Currently, labs do not have clear proposals for how they will monitor for dangerous capabilities, and by default appear poised to scale up potentially dangerous models without careful investigation of the risks.

1a. What kinds of topics should AI accountability mechanisms cover? How should they be scoped?

AI accountability mechanisms must anticipate dangerous capabilities that today's models are not yet capable of.

We believe that at some point, AI systems will likely be able to do most of what humans can do, including developing new technologies, starting businesses and making money, finding new cybersecurity exploits and fixes, and much more. Once AI systems are performing these functions on a large scale they could become very difficult, if not impossible, for humans to control. If these systems were accidentally working towards ends that were harmful to humanity, and especially if they could replicate and resist attempts to shut them down, they could become a serious danger to society as a whole.

For this reason we believe that AI accountability regimes intended to reliably ensure public safety must incorporate tests for catastrophically dangerous capabilities that current models are largely incapable of but that the next generation of frontier models might be capable of.

1b. What are assessments or internal audits most useful for? What are external assessments or audits most useful for?

If dangerous capability evaluations are performed by external auditors, it is crucial that these auditors be given sufficient technical access. If dangerous capability evaluations are performed by internal auditors, it is crucial that internal auditors have public accountability.

In ARC's view, thoroughly testing a model for a dangerous capability like Autonomous Replication and Adaptation requires in-depth technical work with the AI model being evaluated. For reasons we will explain later in this comment, we believe that a sufficient audit requires the ability to "fine-tune" an AI model for the dangerous capability being tested, and, probably, the ability to conduct tests at checkpoints while an AI model is still being trained. Thus, for an external auditor to perform dangerous capability evaluations, it is critical that they be given access comparable to a red-team inside of a lab developing the AI model itself.

It may be the case that it is difficult for the employees of an AI lab to communicate their technical know-how to external auditors or easily and securely give external parties access sufficient to conduct thorough testing. If this is the case, it may be expedient to adopt a framework of AI accountability in which internal "red-teams" conduct dangerous capabilities evaluations. If such a system of internal auditing is adopted, it is critical that AI labs conducting internal audits state publicly what dangerous capabilities they are evaluating their AI models for, how they are conducting those evaluations, and what actions they would take if they found that their AI models exhibited dangerous capabilities. Public commitments will ensure that the wider society is kept informed of the risks posed by the research in question, confirm that sufficient dangerous capability evaluations are being conducted, and ensure accountability if a lab changes its plans for dangerous capability testing.

In short, if dangerous capability evaluations are performed by external auditors, it is crucial that these auditors be given sufficient technical access. If dangerous capability evaluations are performed by internal auditors, it is crucial that internal auditors have public accountability.

1e. Can AI accountability practices have meaningful impact in the absence of legal standards and enforceable risk thresholds? What is the role for courts, legislatures, and rulemaking bodies?

Legislatures and rulemaking bodies may want to require that AI labs have plans to conduct dangerous capability evaluations before building more capable models, or require that specific dangerous capability evaluations are conducted by external auditors if doing so is feasible.

ARC Evals is hopeful that companies may voluntarily choose to adhere to standards and/or practices that reduce the risk of harm from AI. However, there is no guarantee that this will happen, and we are concerned that commercial incentives will not lead to enough investment in preventing rare, large-scale catastrophes. It could be important for legislators, regulators, etc. to require measurement of potential dangerous capabilities before training and/or deployment of models that are much more capable than the current state of the art.

4. Can AI accountability mechanisms effectively deal with systemic and/or collective risks of harm, for example, with respect to worker and workplace health and safety, the health and safety of marginalized communities, the democratic process, human autonomy, or emergent risks?

Clear, concrete dangerous capability definitions are key to establishing accountability.

Currently, there is no clear widely agreed upon measure of whether a given AI system poses a catastrophic risk to human society, and developing such a measure is a key unsolved area of technical research. If we could develop testable lower-bounds on the degree and kind of capability that would make us seriously worried that an AI system posed a catastrophic risk, this would help for several reasons:

- Standards organizations and governments, when concerned about the safety of a class of models, could make more specific asks or requirements of AI research labs. Rather than blunt instruments like pausing AI research on a class of models as a whole, those advocating for safety could be specific about the criteria that would cause them to trigger a pause and generally be more specific about the types of AI systems that they would like to prohibit the development and deployment of.
- When an AI research lab claimed that it was taking an action in the name of safety, for instance slowing down the pace of compute scaling for large AI systems or adopting more secretive publication norms, other actors (e.g. regulators, competitors) wouldn't have to take the lab's word for it. Research labs inclined to take cautious actions would have widely agreed upon tests that they could use to demonstrate their AI systems were now demonstrating early signs of dangerous behaviors.
- Clear definitions of what capabilities qualify as dangerous would make AI research labs unencumbered to freely pursue additional technical advances. General research into AI advances

could proceed more rapidly, in the knowledge that dangerous capabilities would be noticed if AI systems became capable of them.

11. What lessons can be learned from accountability processes and policies in cybersecurity, privacy, finance, or other areas?

For AI accountability, inspiration should be taken from domains with high access / high discretion auditing regimes.

Looking at US domestic regulations in other complex, high-stakes technological domains reveals that the degree of *access* and *discretion* granted to an auditor is critical:

- *Access* refers to the extent of the auditor’s power to inspect company information or property.
- *Discretion* refers to the auditor’s ability to deviate from a rigid “checklist” of standards and instead experimentally investigate company information or property for problems that are harder to foresee.

An auditor cannot use their discretion to investigate or red-team a system unless they have enough access to observe and experiment with it. But high access does not necessarily entail high discretion. So there are three possibilities: high access / low discretion, high access / high discretion, and low access / low discretion. Looking at other case-studies, we think that a high access / high discretion regime is best suited to generative AI.

The New York Fed’s banking supervision system is one such *high access, high discretion* regime:

- There is a New York Fed supervisory team embedded within each major bank in New York. This degree of physical access is effective: for instance, a meta-analysis of several studies finds that the closure of such physically-embedded offices resulted in banks being exposed to greater risk.¹
- The teams have strong discretion, for example undertaking informal explorations of one-off issues that seem concerning to auditors.² Similarly, DeepMind researchers themselves believe that AI auditors should be able to “purposefully search for” concerning capabilities.³

The US biosecurity regime, the Federal Select Agent Program (FSAP), is *high access* but *low discretion*:

- FSAP officials also have a high degree of access: for example, they are empowered to “Inspect... entities that possess, use, or transfer select agents.”⁴ This level of oversight over lab personnel seems to have improved biosafety relative to non-FSAP-regulated labs.⁵

¹ Hirtle and Kovner (2021): 11

² Eisenbach et al. (2017): 71

³ Shevlane et al. (2023): 11

⁴ Federal Select Agent Program (2023): see bullet point list

⁵ Richards, Pompei and Anderson (2014): see abstract and p. 4

- However, FSAP officials probably do not have sufficient discretion to prevent future risks. FSAP regulations cover only existing pathogens, not “potential pandemic pathogens” that could be enhanced using modern synthetic techniques. But because “even informed experts might disagree on whether a... pathogen is an enhanced potential pandemic pathogen”⁶, rigid checklist-type standards are obviously inadequate. Verification of potential dangers in AI systems will often be similarly complex,⁷ requiring strong discretion on the part of auditors.

In comparison, *low-access, low-discretion* regimes seem less well-adapted to high-risk domains like AI:

- The International Atomic Energy Agency’s (IAEA) civil nuclear safety standards, in contrast to its nuclear weapons verification program, provides only voluntary checklist-type standards without inspection.⁸ In the wake of the Fukushima-Daiichi nuclear disaster, a prominent *Nature* paper argued that the IAEA’s civil nuclear safety standards should be enforced by law.⁹
- The Biosafety in Microbiological and Biomedical Laboratories standards are voluntary standards followed by most biolabs. But, as noted above, FSAP standards result in higher safety relative to the baseline.¹⁰ 41% of biodefense researchers themselves prefer clear regulations to the BMBL.¹¹

In the AI industry, leading AI labs themselves favor strong auditor access along the lines of the New York Fed’s banking supervision system or the IAEA’s nuclear weapons verification program. OpenAI lab leaders in May called for a IAEA-style regulator “that can inspect systems, require audits, test for compliance with safety standards”, and so on.¹² And as noted above, DeepMind researchers also favor a high-discretion auditing regime.

15d. Since the effects and performance of an AI system will depend on the context in which it is deployed, how can accountability measures accommodate unknowns about ultimate downstream implementation?

It is insufficient to test whether an AI system is capable of dangerous behavior under the terms of its intended deployment. Thorough dangerous capabilities evaluation must include full red-teaming, with access to fine-tuning and other generally available specialized tools.

We believe this is an essential question, and to answer it we’ll reference several technical concepts that it might help to first define for this and future questions:

⁶ Epstein (2023): see the section “Utilizing existing legislation”

⁷ For example see Shevlane et al. (2023): 10

⁸ IAEA (2023a): see the section “How are they applied?”

⁹ Bunn and Heinonen (2011): 1581

¹⁰ Richards, Pompei and Anderson (2014): see abstract and p. 4

¹¹ Sutton (2009): 225

¹² Altman, Brockman and Sutskever (2023): see the section “A starting point”

- **“Frontier” AI Systems:** Approximately, this is the set of models that are the most powerful and general (capable at a wide range of tasks), primarily because they were trained with the most computing power.
- **Fine-Tuning:** The practice of taking a large and generally capable model trained on a large and domain-agnostic body of data, for instance a large language model, and then training the model a bit more on a specific task, to generate responses or take actions in a specific format, or on additional knowledge bases of data covering a specific topic. Practices of this type have shown success at eliciting useful behavior from AI systems that were previously knowledgeable but difficult for laypeople to use, or capable at reasoning, but missing information about a specific discipline. As an example, “ChatGPT”, which captured the public’s imagination, was a version of a text-completion model, GPT-3, that had been specifically fine-tuned to give responses as though it were a virtual assistant.
- **“Model Architecture and Parameters”:** This will be used as a shorthand to refer to both information about the design of an AI system (e.g. its number of parameters) and a model’s “weights”, or the actual coefficients of the neural network itself (i.e. the values that are actually plugged into a neural network to let it execute the complex functions that make up its operation).

Many of the preventative measures against offensive, biased, or malicious behavior towards users that AI developers might make use of, for instance fine-tuning for harmlessness and usage restrictions, are insufficient assurance of safety, since these protections might be engineered or fine-tuned away if a model’s architecture and parameters were stolen or leaked. For this reason, we believe it is important that any auditor seeking to assess AI systems for dangerous capabilities is given, by the developer being audited, whatever tools, resources, and fine-tuning capability an incautious actor with access to stolen or leaked model architecture and parameters might make use of. To faithfully evaluate models with all of the advantages that a motivated outsider would have with access to a model’s architecture and parameters, auditors must be given resources that enable them to simulate the level of access that would be available to a malign actor if the model architecture and parameters were stolen.

Auditing procedures must not substantially increase the chance of a model’s architecture and parameters being leaked or stolen.

Performing such an audit requires a very high level of trust and security on the part of the auditor, so as to ensure that the process of auditing a model for dangerous capabilities does not itself increase the risk that the model architecture and parameters are stolen. For this reason, it is critical that auditors maintain exceptional information security programs, so that auditing adds no additional risk of an AI model’s architecture and parameters being unintentionally leaked or stolen.

16b. How should AI audits or assessments be timed? At what stage of design, development, and deployment should they take place to provide meaningful accountability?

Evaluate AI systems for dangerous capabilities during large-scale training, not only in advance of public deployment.

While testing frontier AI systems for dangerous capabilities is only one form of accountability, we believe it is essential that auditing for dangerous capabilities in particular occurs during the model development and data training period of AI system development. While it may be tempting to only require testing for dangerous capabilities after training is complete and a decision is being made about whether an AI system should be deployed in products and services available for public use, we believe this is insufficient. The reasons are several:

- Once a model is trained at great expense, there could be substantial commercial or political pressure on a lab to deploy it, even if doing so is against the lab's best judgment. The pressure to deploy may cause a breakdown in the lab's normal pre-deployment testing procedure.
- Even if no intentional public deployment of a model occurs, its model architecture and parameters may be stolen or leaked. For instance, a model that was sufficiently capable could become a tempting target for a foreign adversary or, more generally, an Advanced Persistent Threat (APT) willing to spend tens of millions of dollars to try and obtain it. This actor, once in control of a model, may deploy it in ways that the lab developing it did not.
- The Autonomous Replication and Adaptation (ARA) capability that we advocate testing for is, definitionally, a capability that enables AI systems to survive and spread on their own without human support. If a model were capable of this, it would be hard (due to the opaque nature of deep neural networks) to have a sense of whether the model might spontaneously exhibit this behavior.
- Even before public or commercial deployment, a model's coming into existence via training can have an effect on the rest of the world. Most obviously, rumors, screenshots, etc. about a trained model can prompt reactions from other labs.
- Frontier AI models have sometimes been shown to experience large discontinuous jumps in capability at certain tasks during training runs, and it has sometimes been difficult to predict when specific AI capabilities will emerge. If a lab waits to test for dangerous capabilities until after a large amount of training computing resources have been expended, the lab risks substantially overshooting any given dangerous capability threshold. Testing periodically during training, on the other hand, creates the opportunity to notice improving performance on dangerous capability benchmarks with finer resolution. Risk management will be easier if we are able to detect a model's capacity for dangerous behaviors like Autonomous Replication and Adaptation soon after the model first demonstrates competency at this set of behaviors.